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Stochastic Optimization of Spatial Throughput in Wireless Networks under Determinantal Scheduling

Project Context and Motivation

Domain: Wireless Telecommunications and MAC Layer Protocols.

Recap: Maximizing the number of successful data streams before interference crashes the network.

Objective: Maximizing the number of successful data streams before interference crashes the network.

Why uncertainity is important?

Real-world networks operate in uncoordinated, unlicensed bands (like Wi-Fi ISM bands). User locations and physical signal fading are inherently unpredictable.

Project Objective and Scope Main Goal:

- M3 placed all transmitters at fixed coordinates using numpy.meshgrid
- Two layouts tested: square grid and hexagonal grid
- Produced a bell curve with a clear optimal density lambda*

Key Limitations:

- Fails to represent the chaotic clustering of real human behavior.
- Highly susceptible to "boundary effects" in bounded simulations.
- Does not account for dynamic MAC layer contention.

Project System Overview Step-by-step system:

- **End-to-End Pipeline:**

  Spatial Geometry Generation → Distance Calculation → Rayleigh Fading Application → SINR Evaluation → Throughput Output.

- **Major Components:**
  - The Environment: Transmitters, Receivers, and physical boundaries.
  - The Physics Engine: Signal decay and stochastic interference.

- **Decision Rule:**

  Operates at the very end; if SINR > T, the transmission is marked successful.

Sources of Uncertainty in the Project

- **Source 1 (Physics):**

  Multipath Rayleigh fading causing signals to randomly degrade.

- **Source 2 (Spatial):**

  Uncoordinated user arrivals forming chaotic clusters.

- **Why Deterministic is Insufficient:**

  A rigid grid assumes the network architect has total control over where devices are located, which is impossible in modern ad-hoc or Wi-Fi networks.

Key Random Variables and Their Roles

- Fading (H): Modeled with an Exponential Distribution. Represents signal degradation.
- Aggregate Interference (I): The stochastic sum of all background noise.
- NEW - Spatial Coordinates (X, Y): Modeled using Point Processes.

  Replaces the fixed M3 arrays. Represents where active transmitters are

  located at time t.

Randomized Algorithm Design

- **The Decision Rule:** How do we decide which uncoordinated nodes get to transmit?
- **Algorithm 1 - ALOHA (PPP):** Nodes transmit entirely at random (coin flip).
- **Algorithm 2 - CSMA (Listen-Before-Talk):** Mimics 802.11 Wi-Fi. Nodes sense the medium and back off if a neighbor is transmitting.
- **Algorithm 3 - DPP (Base Paper):** Uses probability matrices to mathematically enforce spatial repulsion (social distancing) among active nodes.

Probabilistic Reasoning Behind the Algorithm

- **The Covariance Proof (E[XY]):** Let X = Active Transmitters, Y = Successful Links. They have a negative correlation ({Cov}(X,Y) < 0) because higher density causes interference to choke the network.

- **Why Randomization Helps:** Controlled randomization (like CSMA and DPP) manages MAC layer contention. By actively preventing nodes from transmitting too close to one another, we mathematically reduce the Aggregate Interference (I).

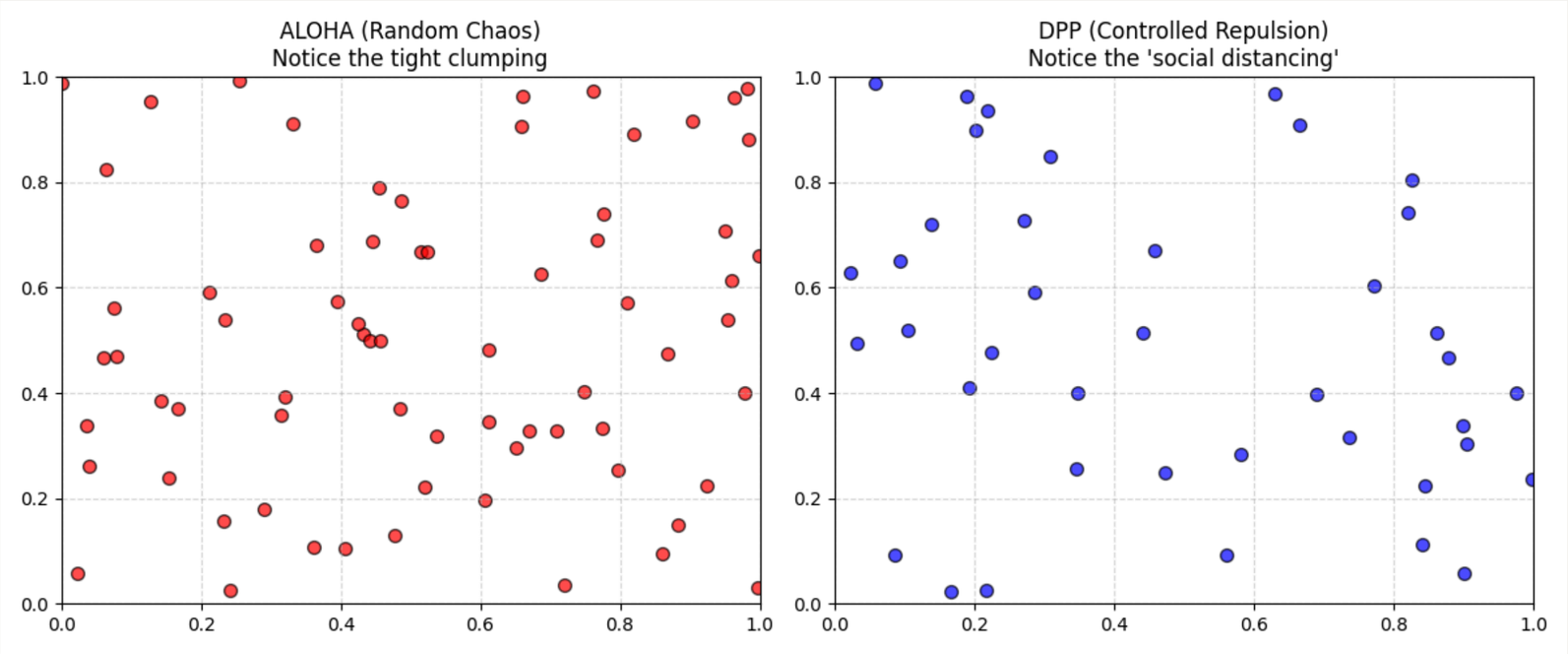
Implementation of the Randomized Strategy

- **Integration:** We swapped the numpy.meshgrid deterministic generator for randomized Point Process generators.
- **Code Structure:** We directly imported the base paper's Python scripts (funSimSimpleDPP.py) for rigorous eigenvalue calculation.
- **Workflow:** Generate random map --> Apply repulsion algorithm --> Feed resulting coordinates into the exact same M3 physics engine.

Evaluation Metrics

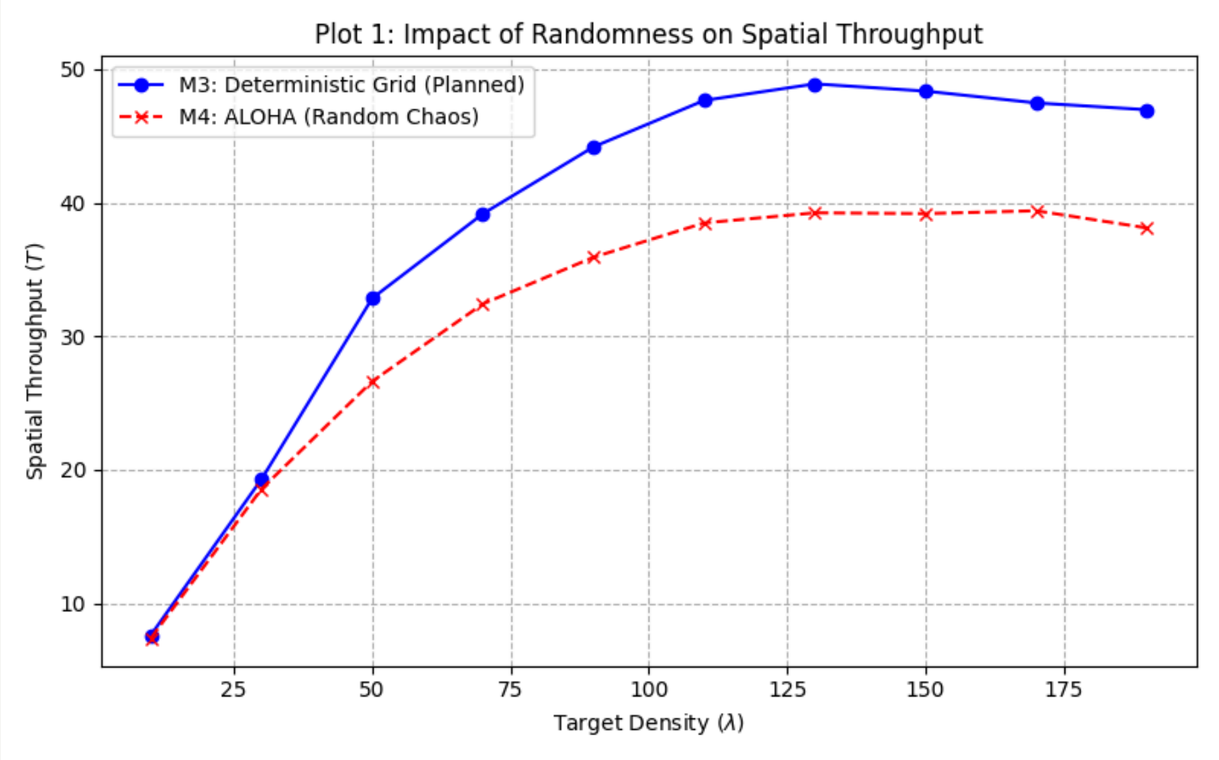
- **Metric 1:** Spatial Throughput **(T)**.
- **Metric 2:** Coverage Probability (Pcov).
- A**lgorithm 2 - CSMA (Listen-Before-Talk):** Mimics 802.11 Wi-Fi. Nodes sense the medium and back off if a neighbor is transmitting.
- **Why Appropriate:** These metrics directly answer the problem statement. They prove whether an algorithm successfully managed interference or allowed the network to crash..

Comparative Results



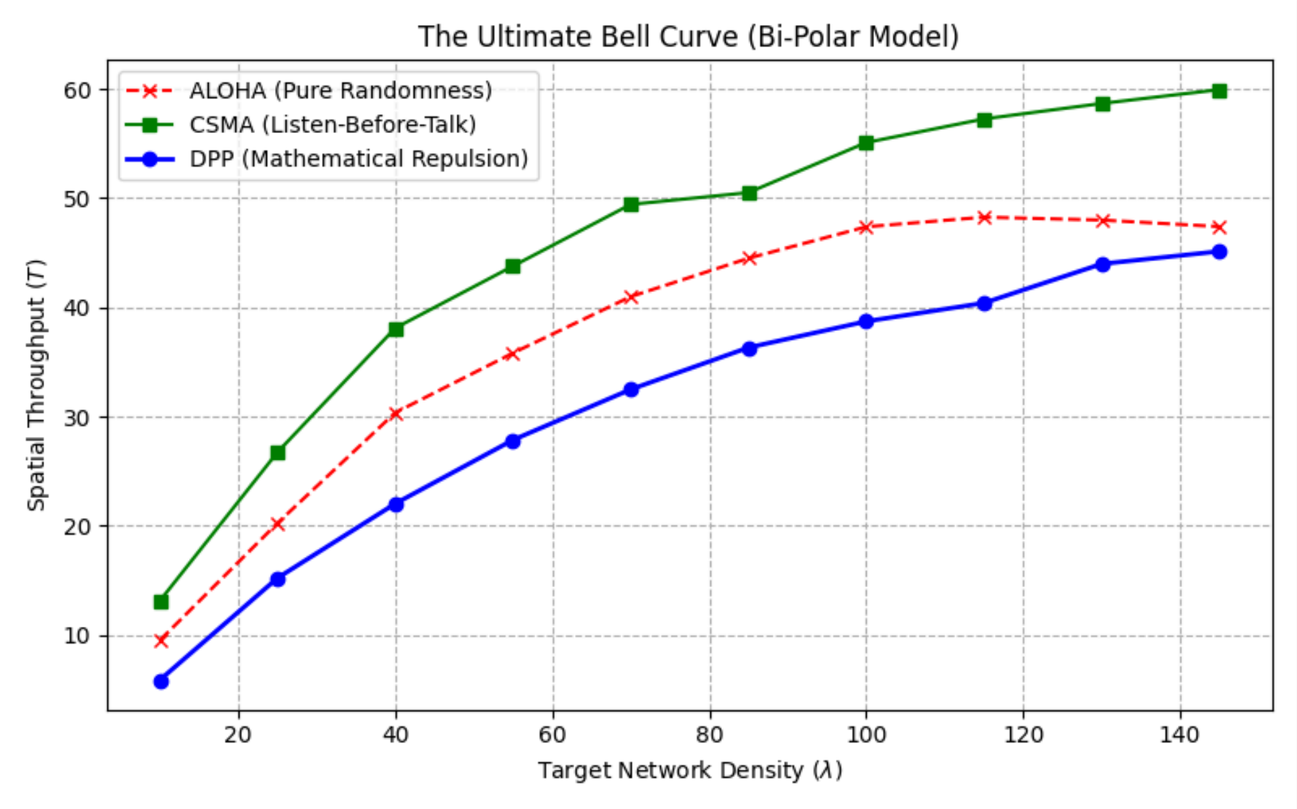
- **Observation:** Aloha Vs DPP Spread

Comparative Results



- **Observation:** Uncontrolled random chaos (ALOHA) performs significantly worse than the deterministic grid.

Comparative Results



- **Observation:** Network Density Vs Throughput in different randomized algorithms

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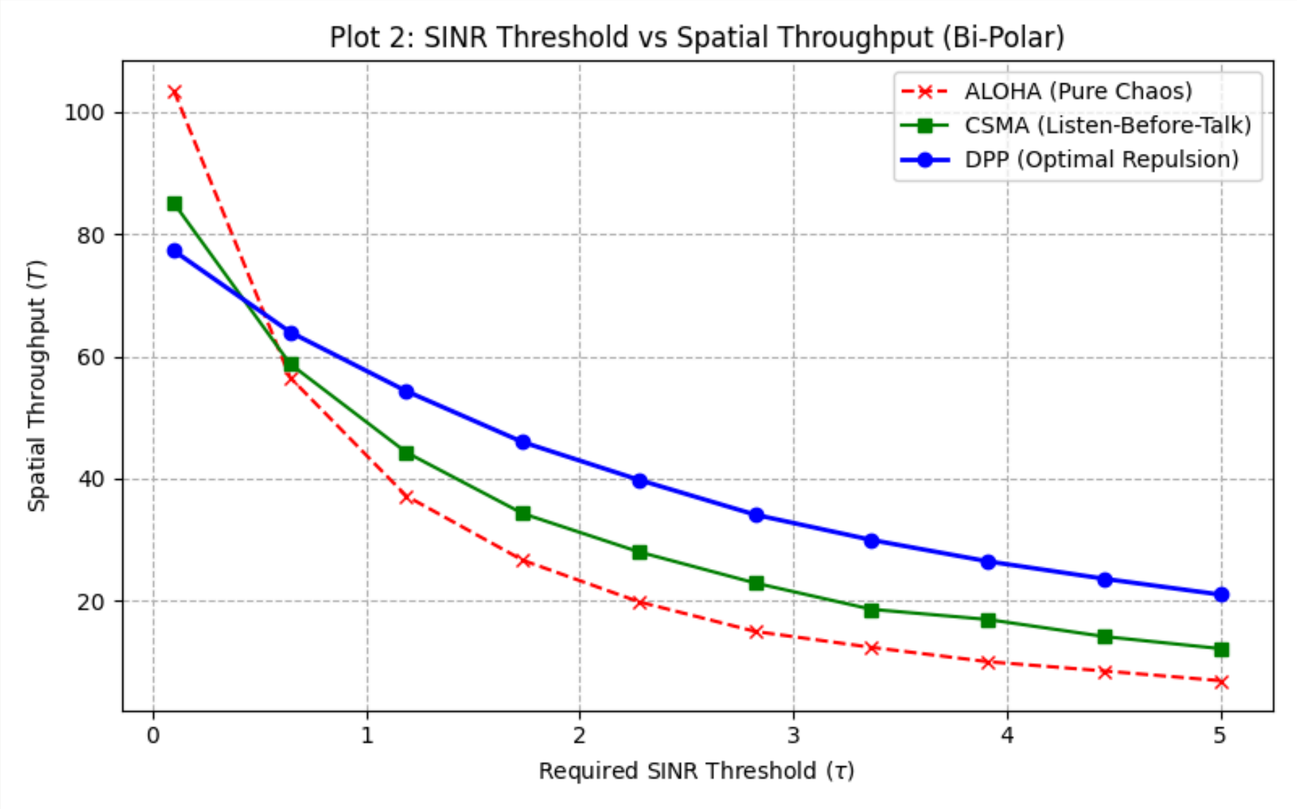
Comparative Results



- **Observation:** When comparing randomized algorithms, DPP achieves the highest throughput by mathematically organizing the chaos.

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Comparative Results

```
Running Monte Carlo Simulation to calculate E[XY]...

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PROFESSOR'S EXPECTATION OPERATOR PROOF
==============================================
E[X]  (Avg Active Transmitters) = 103.47
E[Y]  (Avg Successful Links)    = 40.42
E[XY] (Joint Expectation)       = 3931.68
----------------------------------------------
Covariance Cov(X, Y)            = -250.71
==============================================

CONCLUSION: Negative Correlation Confirmed.
As the number of active transmitters (X) increases, the
resulting interference causes successful transmissions (Y) to drop.
```

- **Observation:** Calculate Cov(X,Y) , Avg Active Transmitters and Avg succesful links and calulating Joint Expectation, E[XY]

Key Insights

- **Insight 1:** Pure randomness (ALOHA) is terrible for telecommunications due to clumping and unchecked interference.
- **Insight 2:** The E[XY] negative correlation proves that simply adding more active transmitters will eventually break the network.
- **Insight 3:** Modern MAC protocols require controlled randomness (Collision Avoidance/Repulsion) to maintain fairness and maximize throughput in uncoordinated environments.